Additional Implementation Details and Experiments

A. Datasets and Experimental Setup Details

In the experiments, we compare the performance of the on four datasets, Google Speech Commands (GSC) [1], SoClas [2], AFPILD [3] and AV16.3 dataset [4]. These datasets cover different common robotics tasks including speaker localization, pedestrian detection and acoustic event localization. This section gives a detail description on the setting of each dataset.

GSC Dataset. To train the proposed model, we construct a simulated acoustic dataset using speech signals from the Google Speech Commands (GSC) dataset as sound sources. Room acoustics are simulated using the Pyroomacoustics [5] toolkit with the image-source method. The simulated environment consists of a $7 \times 7 \times 3$ m room with a sampling rate of 16 kHz, and additive noise is added with a signal-to-noise ratio (SNR) of 30 dB. The sound source is randomly positioned around the microphone array with a distance ranging from 0.5 m to 2.0 m and an azimuth angle uniformly distributed between 0° and 359° with a resolution of 1°. For each azimuth angle, 100 samples are generated for training and 25 samples are generated for testing, resulting in a uniformly distributed angular dataset for supervised direction-of-arrival (DOA) estimation.

AV16.3 Dataset. AV16.3 is a widely used benchmark dataset for speaker localization and tracking in indoor environments. Following the AV3T [6] preprocessing pipeline, the raw recordings are processed and the dataset is divided such that sequences *seq8* and *seq18* are used for testing, while the remaining sequences are used for training. This dataset is used to evaluate the generalization ability of the proposed method under multi-speaker conditions.

SoClas Dataset [2]. The SoClas dataset contains acoustic recordings of various sound events designed for joint sound localization and classification. In our setup, the microphone array is placed at the center of the room while the sound source is positioned at a fixed distance of 1.5 m. The azimuth angle ranges from 1° to 360° with a 5° interval. For each direction, 80% of the samples are randomly selected for training and the remaining 20% are used for testing in the experiments. This dataset evaluates the robustness of the proposed method for event-level sound localization.

AFPILD Dataset. The AFPILD dataset focuses on pedestrian localization using footstep sounds recorded by a microphone array and a LiDAR sensor. We follow the official training and testing split provided by the dataset authors and conduct experiments using the cloth covariant configuration (`covariant_type = 'cloth'`) according to the standard evaluation protocol. This dataset reflects a realistic robotic perception scenario for surrounding pedestrian localization.

It is worth noting that the SoClas and AFPILD datasets do not explicitly provide the world coordinate system, microphone coordinate system, or corresponding calibration files. As a result, traditional DOA estimation methods may exhibit systematic angular offsets when applied directly to these datasets.

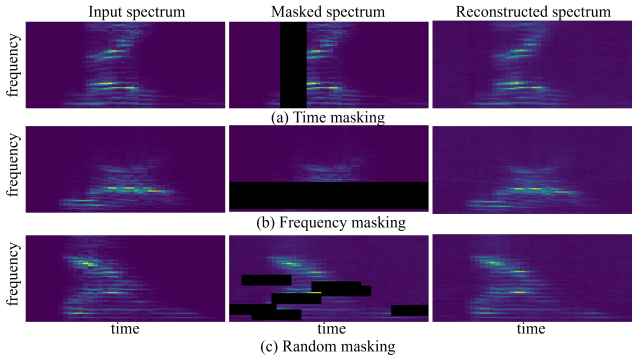

Fig. 1. Masking strategy used in the proposed SSCL module and the corresponding reconstructed samples.

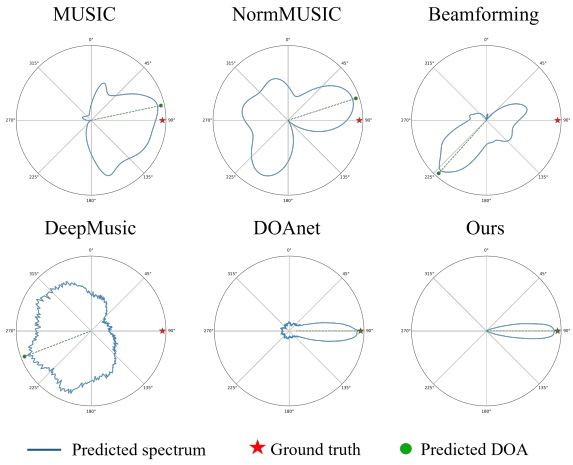

Fig. 2. The predicted spectrum for different methods under 0db noise.

To ensure a fair comparison, we estimate the optimal angular offset between the predicted DOA and the ground truth during evaluation and report the results using the offset that yields the best performance for traditional methods.

B. Training Details

The proposed framework is trained in two stages: (1) the Self-supervised Spatial Correlation Learning (SSCL) stage and (2) the supervised NeuralMUSIC DOA estimation stage.

In the SSCL stage, the model is trained for 400 epochs using the Adam optimizer with a CosineAnnealingLR learning rate scheduler, and a batch size of 256. This self-supervised pretraining encourages the network to learn geometry-consistent spatial representations from multi-channel audio, which improves the robustness and generalization capability for subsequent sound source localization tasks.

In the supervised NeuralMUSIC DOA estimation stage, the network is optimized using the Adam optimizer with an initial learning rate of 1×10^{-3} for 100 epochs. The balancing parameter

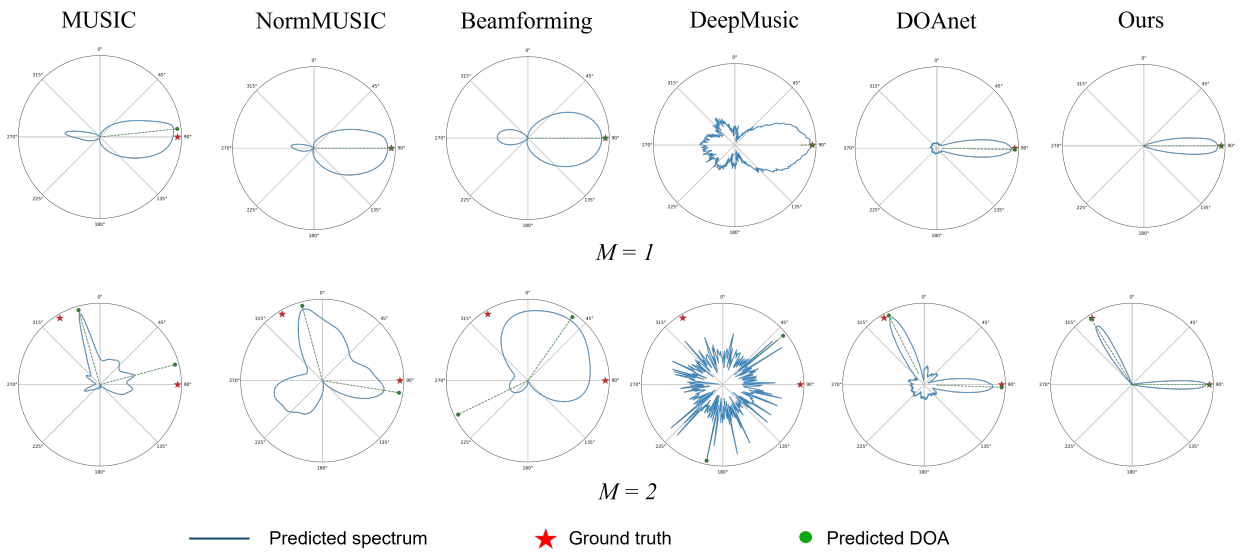

Fig. 3. Predicted DOA spectra of different methods under varying numbers of sound sources.

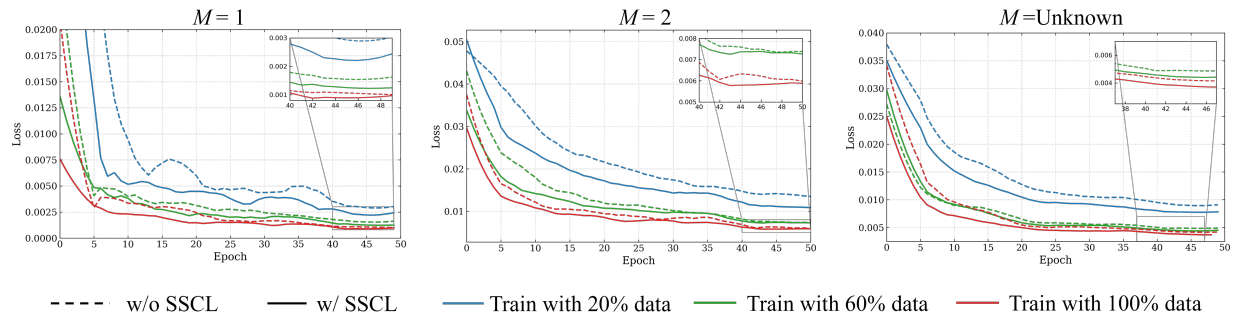

Fig. 4. Training loss curve under different training data ratio with (w/) and without (w/o) SSCL

λ is set to 0.5 and the batch size is 32. All deep learning models are implemented in PyTorch and trained on a NVIDIA RTX 4090 GPU.

C. DOA Estimation Results Visualization

To provide a qualitative comparison of different approaches, we visualize the DOA spectra generated by representative methods. Fig. 3 illustrates the localization results under different sound source number settings. It can be observed that the proposed method produces the most accurate and clean DOA spectra, with sharp peaks located around the ground-truth directions across different scenarios.

In contrast, although traditional subspace-based methods can achieve accurate localization in the single-source case, their performance degrades significantly when multiple sources are present. DOAnet [7] also demonstrates competitive performance under both single- and multi-source settings. However, compared with the proposed method, its predicted DOA spectra are less precise, exhibit stronger noise artifacts, and contain more spurious side lobes.

Fig. 2 further presents the localization performance under low signal-to-noise ratio conditions (SNR = 0 dB), where additive white Gaussian noise (AWGN) is introduced into the simulated environment. A similar trend can be observed: the performance of classical methods deteriorates significantly and produces unreliable spectra. In contrast, the proposed method remains robust, generating the cleanest spectral responses and providing the most accurate DOA estimates.

D. Effectiveness of SSCL

In order to better exploit the unlabeled data collected during robot operation, we introduce the Self-supervised Spatial Correlation Learning (SSCL) module. The masking strategy used in SSCL and examples of reconstructed samples are illustrated in Fig. 1.

To evaluate whether SSCL benefits the downstream DOA estimation task, we compare the training curves under different data ratios and source-number settings, as shown in Fig. 4. It can be observed that models initialized with SSCL consistently start from a lower initial loss, indicating a better initialization. Moreover, SSCL leads to faster convergence and achieves lower final training losses compared with models trained from scratch. This trend remains consistent across different training data ratios and source-number conditions.

These results demonstrate that SSCL provides an effective initialization and enables the network to learn more stable

TABLE I
MEAN ANGULAR ABSOLUTE ERROR (MAAE IN °) OF DIFFERENT METHODS ON THE **SLoClas** AND **AFPILD** DATASETS UNDER VARYING TRAINING RATIOS. LOWER IS BETTER.

Method	**SLoClas**					**AFPILD**				
	100%	**80%**	**60%**	**40%**	**20%**	**100%**	**80%**	**60%**	**40%**	**20%**
MUSIC [8]	8.93	8.93	8.93	8.93	8.93	16.02	16.02	16.02	16.02	16.02
NormMUSIC [9]	5.96	5.96	5.96	5.96	5.96	15.90	15.90	15.90	15.90	15.90
Beamforming [10]	7.50	7.50	7.50	7.50	7.50	21.49	21.49	21.49	21.49	21.49
TOPS [11]	16.29	16.29	16.29	16.29	16.29	64.40	64.40	64.40	64.40	64.40
CSSM	22.88	22.88	22.88	22.88	22.88	30.16	30.16	30.16	30.16	30.16
WAVES	25.18	25.18	25.18	25.18	25.18	53.17	53.17	53.17	53.17	53.17
FRIDA [12]	26.11	26.11	26.11	26.11	26.11	56.71	56.71	56.71	56.71	56.71
CRNN [3]	3.37	3.87	4.72	4.78	6.71	10.55	10.82	**10.85**	11.97	22.69
Transformer [13]	1.77	3.31	3.56	4.51	5.32	11.16	11.27	12.23	12.84	14.72
DOANet [7]	3.60	3.93	3.96	5.06	6.41	12.78	13.09	13.73	14.32	20.00
DeepDAE [14]	3.87	4.19	5.08	6.62	8.66	17.29	19.48	21.98	23.00	31.63
DeepMusic [15]	5.80	6.31	8.76	12.07	14.01	14.98	15.04	15.87	16.73	22.86
DA-Music [16]	4.87	5.79	6.60	8.28	10.29	15.72	17.40	17.65	20.58	29.81
Ours	**2.91**	**3.13**	**3.33**	**3.96**	**5.15**	**10.42**	**10.48**	11.22	**11.95**	**13.41**

spatial representations, thereby improving both convergence efficiency and overall localization performance in downstream sound source localization tasks.

E. Data Efficiency Testing

By incorporating physical prior knowledge and effectively leveraging unlabeled data through spatial correlation learning, the proposed method achieves superior data efficiency compared with purely deep-learning-based approaches. We further evaluate the proposed method on two real-world datasets. The quantitative results are summarized in Table I, which further demonstrate the robustness and generalization capability of our approach in practical acoustic environments.

On the **SLoClas** dataset, the proposed method achieves the lowest MAAE across all training ratios. Even when the training data is reduced to only 20%, our method maintains a relatively low error of 5.15°, while most deep-learning-based methods exhibit noticeable performance degradation. On the more challenging **AFPILD** dataset, which involves real-world pedestrian localization scenarios, our method also consistently outperforms existing approaches. In particular, when only 20% of the training data is used, our method achieves an MAAE of 13.41°, showing strong robustness under limited training data conditions.

Overall, these results demonstrate that the proposed framework achieves superior localization accuracy while maintaining strong robustness and data efficiency across different acoustic environments.

F. Generalization Test on Sparse Training

To further evaluate the generalization capability of the proposed method, we conduct a more challenging experiment in which the models are trained only on sparsely sampled DOA angles while being evaluated on the full DOA range from 0° to 360°. Specifically, during training, the DOA space is sampled at coarse angular intervals (e.g., 10°, 20°, ..., 350°), whereas the testing phase covers the entire angular space with a 1° resolution.

TABLE II
MAAE (°) OF DIFFERENT METHODS TRAINED USING SPARSE ANGULAR SAMPLING (5° AND 10° INCREMENTS) AND EVALUATED ON THE FULL DOA RANGE.

Methods	**Sparse 5° Training**	**Sparse 10° Training**
MUSIC	3.96	3.96
NormMUSIC	1.97	**1.97**
Beamforming	2.47	2.47
TOPS	5.14	5.14
WAVES	17.04	17.04
FRIDA	13.01	13.01
CRNN	7.99	9.07
Transformer	3.13	7.55
DOANet	3.22	5.77
DeepDAE	7.57	5.99
DeepMusic	3.38	14.68
DA-Music	6.91	12.14
Ours	**1.89**	2.35

The quantitative results are summarized in Table II. It can be observed that the proposed method maintains accurate localization performance even when trained with sparse angular supervision. In particular, our method achieves the lowest MAAE under both sparse 5° and 10° training settings. These results demonstrate that the proposed approach possesses strong interpolation capability and superior generalization compared with existing methods.

G. Future Work

The proposed NeuralMUSIC framework achieves strong performance in robot sound source localization by integrating data-driven spatial representation learning with classical subspace-based signal processing. By combining physical priors with self-supervised spatial correlation learning, the method demonstrates improved robustness and data efficiency across both simulated and real-world datasets.

Despite these advantages, several limitations remain. First, although the proposed method shows promising generalization ability, it may still underperform traditional signal-processing methods when the training data are extremely limited, when the training distribution is highly imbalanced, or when the microphone array configuration differs significantly from the training setup. Future work will investigate incorporating stronger physical constraints and developing geometry-invariant learning mechanisms to improve cross-array and cross-environment generalization.

Second, further improving data efficiency remains an important direction. Although SSCL helps leverage unlabeled data, the model may still require a certain amount of labeled samples to achieve optimal performance. Future work will explore few-shot learning techniques to enable reliable localization even with only a few labeled samples.

H. Supplementary: Further Discussion on Differences from Existing Hybrid Methods

Several recent works [16]–[19] have explored hybrid frameworks that combine neural networks with classical subspace-based pipelines. While these methods demonstrate promising performance, they still exhibit several limitations that restrict their applicability in robotic audition scenarios.

(1) Array configuration dependency. Some existing approaches are restricted to specific array geometries. For instance, works such as [17], [18] employ Root-MUSIC [20] within their pipeline, which requires a uniform linear array (ULA). Such a constraint limits their applicability to other microphone configurations commonly used in robotic platforms. In contrast, our framework is built upon the classical MUSIC formulation, which can be applied to more general array geometries, making it more suitable for diverse robotic sensing setups.

(2) Lack of broadband frequency modeling. Most existing hybrid methods [16]–[19] are designed primarily for narrowband signals, which are common in radar sensing. However, acoustic signals in robotic applications are inherently broadband. Simply applying narrowband formulations to broadband signals may lead to suboptimal performance because different frequency bands carry varying levels of spatial information. If unreliable frequency components are not properly handled, the resulting pseudo-spectrum may be distorted by noise. To address this issue, we introduce a Frequency Attention Fusion (FAF) module that adaptively reweights frequency bins and emphasizes informative spectral components when constructing the broadband pseudo-spectrum.

(3) Limited validation in real-world environments. Many existing hybrid approaches evaluate their performance primarily in simulated environments. While simulation provides controlled conditions for algorithm development, real-world acoustic environments are often significantly more complex due to reverberation, sensor noise, and environmental interference. Without extensive validation on real-world recordings, the practical effectiveness of these methods remains unclear. In contrast, our method is evaluated on three different robotic audio datasets, providing a more comprehensive assessment of its robustness and practical applicability.

(4) Loss of interpretability and physical structure. Although several hybrid methods combine neural networks with classical algorithms such as MUSIC or Root-MUSIC, these approaches [16], [19] introduce additional neural networks after the classical pipeline to directly regress the DOA. Such additional regression modules effectively transform the overall system back into a black-box model and weaken the physical interpretability provided by the subspace formulation. This design may also reduce the generalization ability inherited from classical signal processing models. In contrast, our framework preserves the classical pseudo-spectrum generation process and introduces only a lightweight frequency attention module to fuse information across frequency bins. This design maintains the physical structure of the MUSIC algorithm while improving robustness.

(5) Differences in pipeline design. Taking DA-MUSIC [16] as an example, its source number estimation module is placed after the eigenvalue decomposition (EVD) stage. However, at this stage some information from the raw input may already be lost, as the estimation relies on the intermediate covariance representation. This may reduce the accuracy of source number estimation. In contrast, our method predicts the source number directly from the input spectral representation, enabling the model to utilize richer spatial information for more reliable estimation.

Furthermore, DA-MUSIC directly regresses the DOA angle. This design introduces two potential challenges. First, defining an appropriate regression loss becomes more complicated when the number of sources is unknown. Second, training stability may be affected because DOA estimation typically involves peak extraction from the pseudo-spectrum, which introduces non-differentiable operations and weak gradients. DA-MUSIC addresses this issue by introducing an additional neural regression module. While effective, this approach further reduces the interpretability of the overall system. In contrast, our method directly outputs the MUSIC pseudo-spectrum and estimates DOA through peak selection. This design preserves the physical meaning of the classical algorithm while maintaining a simple and effective learning pipeline. Moreover, representing localization results as a spatial probability distribution rather than a single precise angle is more consistent with human intuition, as it provides an interpretable confidence over possible directions.

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